

Republic of the Philippines  
**Western Mindanao State University**  
**College of Computing Studies**  
Zamboanga City

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## **WMSU FACULTY LEAVE APPLICATION SYSTEM**

***(WD 123 | CC 104)***

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## I. PROJECT CONTEXT

Currently, *Western Mindanao State University (WMSU)* processes faculty leave applications using manual, paper-based methods. This traditional approach often leads to problems such as lost forms, slow approval times due to physical routing, and mistakes in calculating remaining leave credits. Additionally, faculty members often do not know their current leave balance or the status of their application without visiting the HR office.

To solve these issues, this project introduces the ***Faculty Leave Application System***. This is a secure, web-based platform that moves the entire process online. It replaces manual recording with an automated system that ensures accurate data, faster processing, and better communication between faculty and the administration.

## II. GENERAL OBJECTIVE

To develop a secure **Web-based Faculty Leave Application System** for Western Mindanao State University that allows online leave filing, automatically calculates leave credits, and provides a central digital record for all leave transactions.

### III. SPECIFIC OBJECTIVE

- A. To secure system access by implementing a "Gatekeeper" registration method, ensuring only verified university personnel can create accounts.
- B. To automate leave credit calculations using a "Banking" logic that accurately adds credits based on length of service and adjusts correctly when an employee is promoted.
- C. To simplify the application process with a user-friendly form that automatically calculates the number of leave days (excluding weekends) and requires attachments for specific leave types.
- D. To improve the approval process by allowing Department Heads and Admins to review and decide on applications online, while automatically notifying employees via email.
- E. To streamline reporting by providing a visual dashboard for admins and automatically generating printable PDF leave forms for physical filing.

### IV. SCOPE AND DELIMITATION

#### A. SCOPES

1. **User Management:** Handles two main roles: Employee (Faculty/Staff) and Administrator (HR/Head).
2. **Registration:** Uses a verification system where accounts must be pre-loaded by HR or manually approved by an Admin.
3. **Leave Filing:** Allows applying for Vacation and Sick leaves, complete with date calculations and file uploads.

4. **Credit Computation:** Automatically adds leave credits based on the user's position and time served. It handles promotions safely without calculation errors.
5. **Notifications:** Sends alerts via email and an in-app notification bell when applications are updated.
6. **Reports:** Generates visual charts for the admin dashboard and downloadable PDF forms.

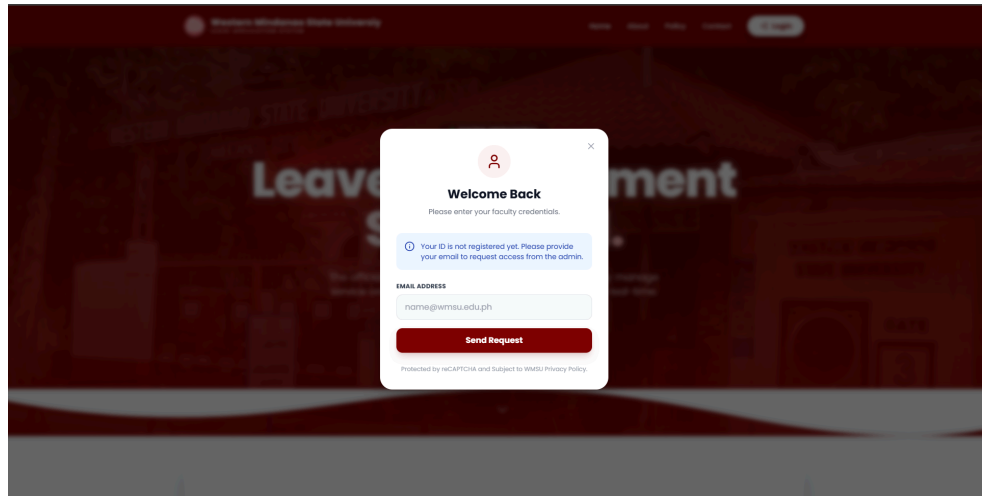
## ***B. LIMITATIONS***

1. **Internet Requirement:** The system is web-based and requires an internet connection to work.
2. **No Payroll Calculation:** The system tracks *leave days* but does not calculate *salary deductions*. It provides the data needed for payroll but does not process the money.
3. **Leave Types:** It is currently set up for standard Sick and Vacation leaves. Special leave types (like Maternity or Study Leave) are not yet included.
4. **No Biometrics:** The system does not connect to fingerprint time clocks; it relies on the data entered by the faculty and HR.
5. **No Replacement:** The system does not hold a schedule of the instructors to be the basis of the replacement of an on-leave instructor.

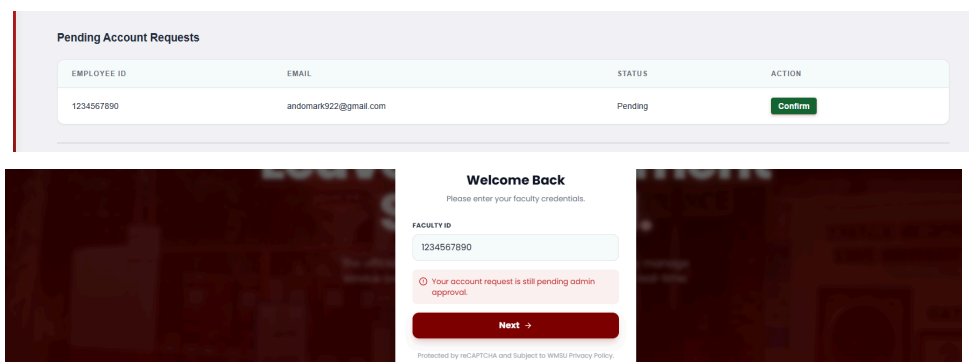
## V. SYSTEM FEATURES

### A. Security & Access

- **Strict Registration:** Public sign-ups are disabled. Users can only register if they have a valid Employee ID and Email that matches HR records.



- **Admin Approval:** New accounts stay "Pending" until an Admin confirms they are real employees.



- **Secure Sessions:** The system automatically logs users out completely to prevent unauthorized access from shared computers

## B. Employee Features

- **File Upload:** Allows users to attach necessary proof, such as

The screenshot shows a web form for submitting a leave application. The form includes the following fields and controls:

- Leave Type:** A dropdown menu with "Sick" selected.
- Start Date:** A date input field showing "17/12/2025".
- Number of Days:** A text input field with the value "3".
- End Date:** A date input field showing "22/12/2025".
- Reason for leave:** A text area containing the text "I'm going to see a doctor with today's appointment."
- Supporting document:** A section with a blue border and dashed outline. It contains the text "Supporting document is optional for sick leave." and an "Upload Document:" label.
- File Selection:** A button labeled "Choose File" and a status "No file chosen".
- Submit:** A red button at the bottom of the form.

**PDF Leave Application Generator:** The system generates a softcopy of the application for printables with the attachment that was submitted.

The screenshot displays a PDF document generated by the system. The document is titled "Leave Application - 600229" and is presented in a two-page layout. The right page contains the following information:

- Header:** Republic of the Philippines, WESTERN MINDANAO STATE UNIVERSITY, Normal Road, Balasan, Zamboanga City 7000. Leave Application System, Date Generated: 2025-12-18.
- Employee Information:**
  - Employee ID: 202403855
  - Full Name: Mark NIA Ando
  - Position: Admin
  - College: College Of Liberal Arts
  - Email: andomark82@gmail.com
- Leave Details (Application ID: 600229):**
  - Leave Type: Sick
  - Total Days: 3.00
  - Start Date: 2025-12-17
  - End Date: 2025-12-19
  - Reason: I&#038;m going to see a doctor with today&#038;s appointment.
- Approval Decision:**
  - Final Status: Pending
  - Decision Date: N/A
  - Approver ID: N/A
  - Approver Remarks: No remarks provided

The left page of the PDF shows a thumbnail of the application form and a list of attachments, including a file named "Supporting document".

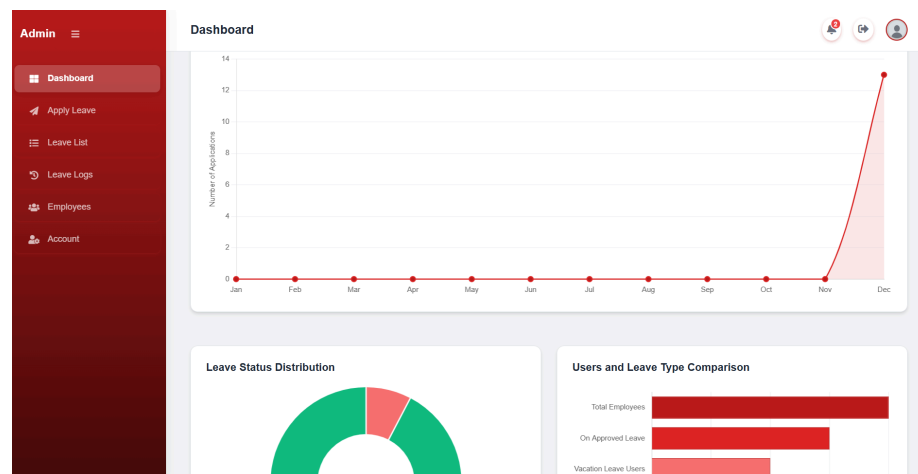
- **Profile View:** Users can see their employment details (which are locked for security) and update their contact info.

The 'Account' page displays the following information for user Mark Ando:

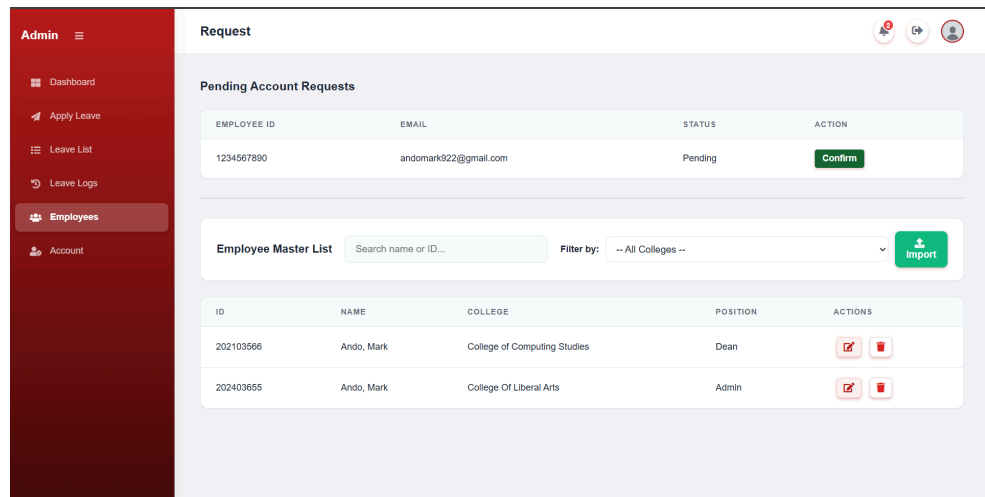
- User Profile:** Mark Ando, Admin, Active Status, Change Password button.
- Employment Details:**
  - EMPLOYEE ID: 202403655
  - DATE HIRED: January 10, 2020
  - COLLEGE: N/A
  - TEACHING STATUS: Administrator
- Personal Information:**
  - FULL NAME: Mark Ando
  - DATE OF BIRTH: July 9, 2005
  - GENDER: Male
- Contact Details:**
  - EMAIL ADDRESS: andomark922@gmail.com
  - MOBILE NUMBER: 09351153438

## C. Admin & HR Features

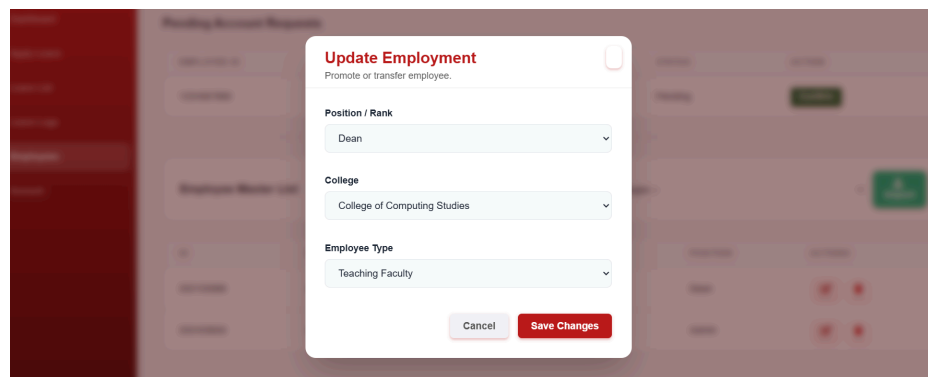
- **Visual Dashboard:** Displays charts showing leave trends, total employees, and current absentees.



- **Employee Master List:** A searchable list of all faculty members that can be filtered by College.

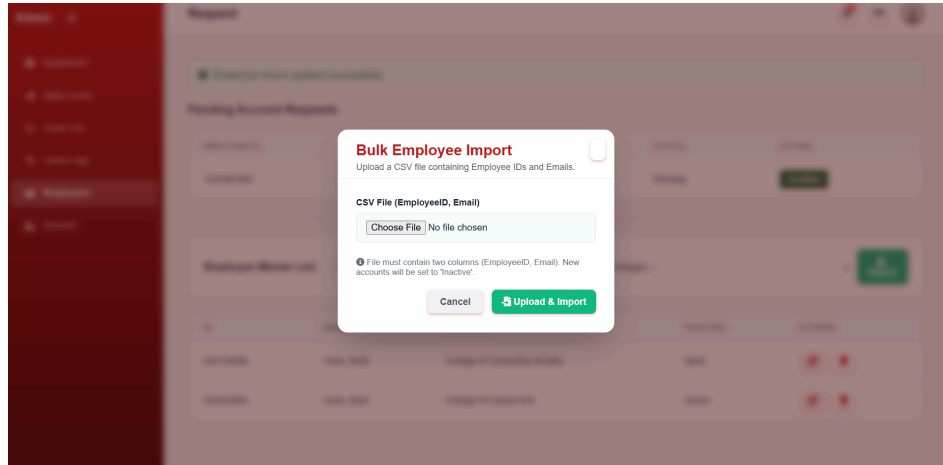


- **Safe Promotion Tool:** A special feature for updating an employee's Rank/Position. It calculates and saves their old credits before switching them to the new, higher accrual rate.





- **Bulk Import:** Allows HR to upload a list of new employees using a CSV file (Excel) to register them quickly.

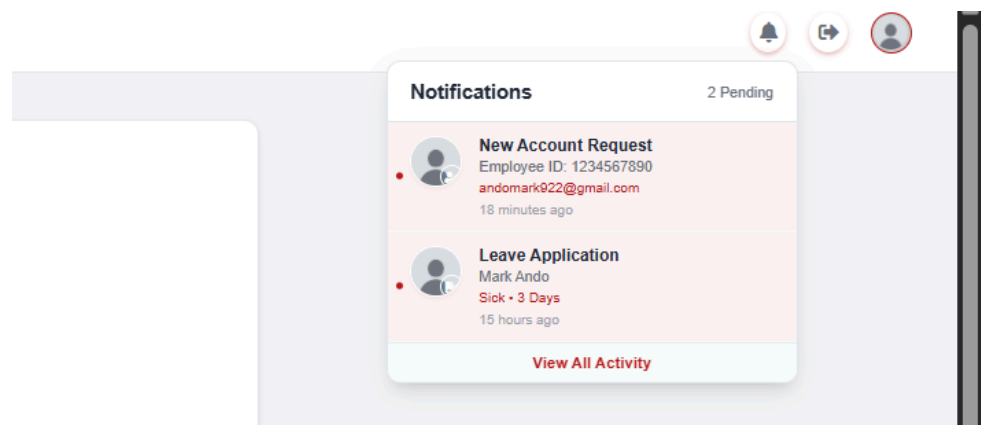


- **Approval Console:** A central place to view pending requests, check attachments, and Approve or Reject applications with comments.

A screenshot of a "LEAVE APPLICATION" form. The form is divided into two main sections: "Personal Profile" and "LEAVE INFORMATION (APPLICATION ID: 600229)".  
**Personal Profile:** Includes a profile picture placeholder and fields for Full Name (Mark Ando), Position (Admin), College (College Of Liberal Arts), Contact Number (09351153438), and Email (andomark922@gmail.com).  
**LEAVE INFORMATION (APPLICATION ID: 600229):** Includes fields for Leave Type (Sick), Total Days (3.00), Start Date (2025-12-17), End Date (2025-12-19), Reason (I&#039;m going to see a doctor with today&#039;s appointment), and Attachment (View File).  
Below these sections is a "Decision" dropdown menu set to "-- Select --", a "Remarks" text area with the placeholder "Enter remarks (optional)", and a red "Submit Decision" button at the bottom.

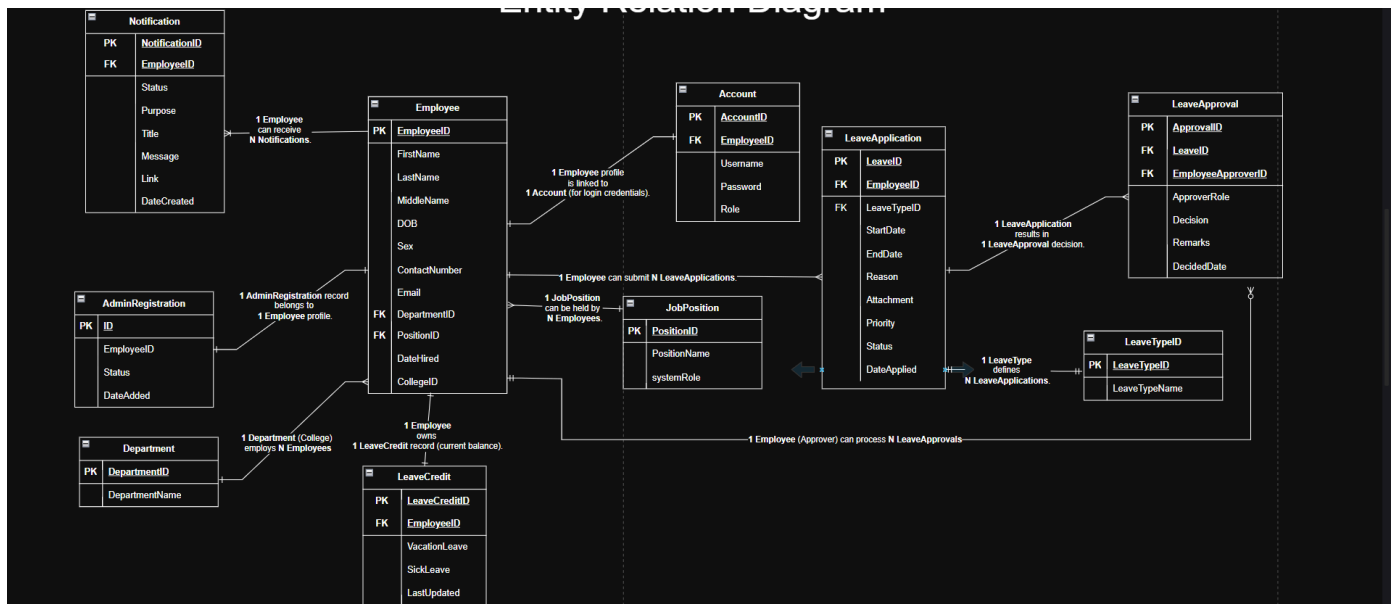
#### D. Core Logic (The "Brains" of the System)

- **Incremental Banking Logic:** Instead of recalculating a user's entire history every time, the system simply adds new credits based on the time passed since the last login.
- **Safe Deduction:** When a leave is approved, the system subtracts days from the balance without messing up the timeline for earning new credits.
- **Automated Alerts:** The system handles all communication automatically, sending emails for account activation, leave decisions, and promotions.



## VI. SYSTEM ANALYSIS AND DESIGN

### A. ENTITY RELATIONSHIP DIAGRAM (ERD)



### B. DATA DEFINITION LANGUAGE (DDL)

```
CREATE TABLE `department` (  
  
    `DepartmentID` INT(11) NOT NULL AUTO_INCREMENT,  
  
    `DepartmentName` VARCHAR(100) NOT NULL,  
  
    PRIMARY KEY (`DepartmentID`)  
);
```

```
CREATE TABLE `jobposition` (  
  
    `PositionID` INT(11) NOT NULL AUTO_INCREMENT,  
  
    `PositionName` VARCHAR(100) NOT NULL,  
  
    PRIMARY KEY (`PositionID`)
```

);

CREATE TABLE `adminregistration` (

    `ID` INT(11) NOT NULL AUTO\_INCREMENT,

    `EmployeeID` VARCHAR(20) NOT NULL,

    `Email` VARCHAR(100) DEFAULT NULL,

    `Status` ENUM('Pending','Active','Inactive','Deleted') DEFAULT 'Pending',

    `DateAdded` DATETIME DEFAULT CURRENT\_TIMESTAMP(),

    PRIMARY KEY (`ID`),

    UNIQUE KEY `EmployeeID` (`EmployeeID`)

);

CREATE TABLE `employee` (

    `EmployeeID` VARCHAR(20) NOT NULL,

    `FirstName` VARCHAR(50) DEFAULT NULL,

    `MiddleName` VARCHAR(50) DEFAULT NULL,

    `LastName` VARCHAR(50) DEFAULT NULL,

    `DOB` DATE NOT NULL,

    `Sex` ENUM('Male','Female') NOT NULL,

    `ContactNumber` VARCHAR(15) DEFAULT NULL,

```
`Email` VARCHAR(100) DEFAULT NULL,  
  
`DepartmentID` INT(11) DEFAULT NULL,  
  
`PositionID` INT(11) DEFAULT NULL,  
  
`DateHired` DATE NOT NULL,  
  
`isTeaching` VARCHAR(10) DEFAULT NULL, -- Consider changing to  
TINYINT(1) or ENUM  
  
`profilePic` VARCHAR(255) DEFAULT NULL,  
  
PRIMARY KEY (`EmployeeID`),  
  
KEY `DepartmentID` (`DepartmentID`),  
  
KEY `PositionID` (`PositionID`),  
  
CONSTRAINT `employee_ibfk_1` FOREIGN KEY (`EmployeeID`)  
REFERENCES `adminregistration` (`EmployeeID`) ON DELETE  
CASCADE ON UPDATE CASCADE,  
  
CONSTRAINT `employee_ibfk_2` FOREIGN KEY (`DepartmentID`)  
REFERENCES `department` (`DepartmentID`) ON DELETE SET NULL  
ON UPDATE CASCADE,  
  
CONSTRAINT `employee_ibfk_3` FOREIGN KEY (`PositionID`)  
REFERENCES `jobposition` (`PositionID`) ON DELETE SET NULL ON  
UPDATE CASCADE
```

);

CREATE TABLE `account` (

    `AccountID` INT(11) NOT NULL AUTO\_INCREMENT,

    `EmployeeID` VARCHAR(20) NOT NULL,

    `Username` VARCHAR(50) NOT NULL,

    `Password` VARCHAR(255) NOT NULL,

    `Role` ENUM('Instructor','Department Head','Dean','Admin') NOT NULL,

    PRIMARY KEY (`AccountID`),

    UNIQUE KEY `Username` (`Username`),

    KEY `EmployeeID` (`EmployeeID`),

    CONSTRAINT `account\_ibfk\_1` FOREIGN KEY (`EmployeeID`)

REFERENCES `employee` (`EmployeeID`) ON DELETE CASCADE ON  
UPDATE CASCADE

);

CREATE TABLE `leavetype` (

    `LeaveTypeID` INT(11) NOT NULL AUTO\_INCREMENT,

    `TypeName` VARCHAR(50) NOT NULL,

    PRIMARY KEY (`LeaveTypeID`),

UNIQUE KEY `TypeName` (`TypeName`)

);

CREATE TABLE `leaveapplication` (

`LeaveID` VARCHAR(10) NOT NULL,

`EmployeeID` VARCHAR(20) NOT NULL,

`LeaveTypeID` INT(11) NOT NULL,

`StartDate` DATE NOT NULL,

`EndDate` DATE NOT NULL,

`NumberOfDays` DECIMAL(5,2) NOT NULL,

`Reason` TEXT NOT NULL,

`Attachment` VARCHAR(255) DEFAULT NULL,

`Priority` ENUM('High','Normal') DEFAULT 'Normal',

`Status` ENUM('Pending','Approved','Rejected') DEFAULT 'Pending',

`DateApplied` DATETIME DEFAULT CURRENT\_TIMESTAMP(),

PRIMARY KEY (`LeaveID`),

KEY `EmployeeID` (`EmployeeID`),

KEY `fk\_leaveapplication\_leavetype` (`LeaveTypeID`),

```
CONSTRAINT `fk_leaveapplication_leavetype` FOREIGN KEY  
(`LeaveTypeID`) REFERENCES `leavetype` (`LeaveTypeID`) ON  
UPDATE CASCADE,
```

```
CONSTRAINT `leaveapplication_ibfk_1` FOREIGN KEY (`EmployeeID`)  
REFERENCES `employee` (`EmployeeID`) ON DELETE CASCADE ON  
UPDATE CASCADE
```

```
);
```

```
CREATE TABLE `leaveapproval` (
```

```
  `ApprovalID` VARCHAR(10) NOT NULL,
```

```
  `LeaveID` VARCHAR(10) NOT NULL,
```

```
  `EmployeeApproverID` VARCHAR(20) NOT NULL,
```

```
  `ApproverRole` ENUM('Department Head','Dean','Admin') NOT NULL,
```

```
  `Decision` ENUM('Approved','Rejected','Pending') DEFAULT 'Pending',
```

```
  `Remarks` TEXT DEFAULT NULL,
```

```
  `DecidedDate` DATETIME DEFAULT CURRENT_TIMESTAMP(),
```

```
  PRIMARY KEY (`ApprovalID`),
```

```
  KEY `LeaveID` (`LeaveID`),
```

```
  KEY `EmployeeApproverID` (`EmployeeApproverID`),
```



```
CONSTRAINT `leaveapproval_ibfk_2` FOREIGN KEY  
(`EmployeeApproverID`) REFERENCES `employee` (`EmployeeID`) ON  
DELETE CASCADE ON UPDATE CASCADE  
);
```

```
CREATE TABLE `leavecredits` (  
  
  `CreditID` INT(11) NOT NULL AUTO_INCREMENT,  
  
  `EmployeeID` VARCHAR(20) NOT NULL,  
  
  `sick` DECIMAL(10,5) DEFAULT 0.00000,  
  
  `vacation` DECIMAL(10,5) DEFAULT 0.00000,  
  
  `lastUpdated` DATETIME DEFAULT CURRENT_TIMESTAMP(),  
  
  PRIMARY KEY (`CreditID`),  
  
  UNIQUE KEY `uk_employee_credit` (`EmployeeID`),  
  
  CONSTRAINT `fk_credits_employee` FOREIGN KEY (`EmployeeID`)  
REFERENCES `employee` (`EmployeeID`) ON DELETE CASCADE ON  
UPDATE CASCADE  
);
```

```
CREATE TABLE `notification` (  
  
  `NotificationID` INT(11) NOT NULL AUTO_INCREMENT,
```

```
`EmployeeID` VARCHAR(20) NOT NULL,  
  
`Status` ENUM('unread','read') DEFAULT 'unread',  
  
`Purpose` VARCHAR(50) NOT NULL,  
  
`Title` VARCHAR(255) DEFAULT 'New Action Required',  
  
`Message` TEXT DEFAULT NULL,  
  
`Link` VARCHAR(255) DEFAULT NULL,  
  
`DateCreated` DATETIME DEFAULT CURRENT_TIMESTAMP(),  
  
PRIMARY KEY (`NotificationID`),  
  
KEY `EmployeeID` (`EmployeeID`),  
  
CONSTRAINT `fk_notification_employee` FOREIGN KEY  
(`EmployeeID`) REFERENCES `employee` (`EmployeeID`) ON DELETE  
CASCADE ON UPDATE CASCADE  
  
);
```